Arnab Ghosh

PhD Student, Electrical and Computer Engineering | University of California, Riverside | Riverside, CA
aghos034@ucr.edu | linkedin.com/in/arnabghosh3107 | github.com/alphalep | Google Scholar

Summary

PhD student in Computational Imaging working across the full vision pipeline, from optics and sensors to learned reconstruction and rendering, with on-device ML experience at OMNIVISION and L&T. Will return to the PhD program after the internship.

Education

University of California, Riverside 2024 – Present
Ph.D., Electrical and Computer Engineering | Advisor: Prof. Vishwanath Saragadam

Rochester Institute of Technology 2021 – 2023
M.S., Imaging Science | Thesis: ShiVaNet, a deep network that restores spatially varying aberration blur, trained with a physics-based blur simulator

Jadavpur University 2014 – 2018
B.E., Instrumentation & Electronics Engineering | First Class with Honours

Publications

1. N. Nezakati, **A. Ghosh**, A. Roy-Chowdhury, V. Saragadam. "CARD: Correlation Aware Restoration with Diffusion." *CVPR*, 2026. [project page] [arXiv]
2. **A. Ghosh**, J. E. Fröch, A. Majumdar, V. Saragadam. "Broadband Content-Adaptive Moiré Meta-spectrometer." Under review, *Communications Engineering*, 2026. [arXiv]
3. **A. Ghosh**, G. Swartzlander. "ShiVaNet: Shift Variant Image Deconvolution using Deep Learning." *IEEE WNYISPW*, 2023. **Best Student Paper Award.** [thesis] [code]

Research Experience

Graduate Student Researcher, UC Riverside Sep 2024 – Present

- **Diffusion restoration under real sensor noise (CVPR 2026).** Modeled spatially correlated rolling-shutter noise from dark-frame covariance and created CIN-D, a 400-image real-noise benchmark, for CARD, a training-free diffusion method with 2–3 dB PSNR gain over DDRM on $2\times$ super-resolution.
- **Moiré meta-spectrometer (first author; with Univ. of Washington).** Turned a rotating Moiré metalens pair into a 405–980 nm spectral scanner and built its Python hardware-in-the-loop stack; a coarse-to-fine adaptive sampling algorithm cuts measurements $\sim 3\times$ (up to $7\times$) and reaches 30 dB $6.7\times$ faster than uniform scanning.
- **Differentiable wave-optics rendering (manuscript in preparation, SIGGRAPH 2027).** New differentiable method for coherent light through lenses, apertures and diffractive elements at wave-optics accuracy and a fraction of standard solvers' cost; validated against wave-optics solvers and commercial lens-design software; GPU PyTorch implementation.

Industry Experience

Software Algorithm Intern, OMNIVISION – *Camera Algorithm Team, Santa Clara, CA* Mar 2024 – Sep 2024

- Built event-camera object detection with binarized neural networks on an in-house edge AI device, reaching 92% accuracy with quantization-aware training.
- Developed memory-efficient, low-power image-processing algorithms for the same device.

Senior Engineer, L&T Technology Services – *Computer Vision Team, Bengaluru, India* Aug 2018 – Jul 2021

- Built a real-time surround-view object-detection alarm system (YOLOv3) and an action-recognition pipeline, both deployed on NVIDIA Jetson TX2.

Technical Skills

ML & Vision	Diffusion Models, Inverse Problems, Image Restoration, Object Detection, Quantization-Aware Training, Binarized Networks, Edge Deployment
Imaging, Optics & Computer Graphics	Computational Imaging & Photography, Deep Optics, Sensor Noise Modeling, Differentiable Rendering, Ray Tracing, Gaussian Beam Propagation, Fourier Optics, Metasurfaces
Math & Programming	Linear Algebra, Numerical Optimization, Probabilistic Estimation; Python, C++, CUDA, MATLAB, PyTorch, OpenCV
Optical Prototyping	Bench Design & Alignment, Imaging & Spectral System Prototyping, Calibration, Hardware-in-the-Loop Acquisition